Huawei Access Node QoS Congestion Event Root Cause Analysis and Resolution SOP

Node: KR-ACC-72 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8090

Description

The failure domain of QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on the Huawei Access Node KR-ACC-72 at KR Puram Access site is primarily attributed to the inherent limitations of the Huawei Access Node's forwarding architecture, specifically the QoS policy configuration and the interface queue configuration. The Huawei Access Node's QoS policy engine relies on the Hierarchical QoS (H-QoS) framework, which categorizes traffic into different classes based on DSCP (DiffServ Code Point) values. However, if the QoS policy is not properly configured, it may lead to QoS queue starvation, causing microbursts and interface tail drops. Furthermore, the Huawei Access Node's interface queue configuration may not be optimized for high-speed interfaces, leading to packet loss and congestion. Additionally, the Huawei Access Node's forwarding architecture may not be equipped with advanced congestion management features, exacerbating the issue. Therefore, it is essential to perform a thorough analysis of the QoS policy configuration and interface queue configuration to identify the root cause of the issue and implement corrective actions to prevent future occurrences.

Pre-requisite Checks

- Ensure that the remote hands are dispatched to KR Puram Access site to perform physical checks and verify the redundant link status.

- Verify that the Huawei Access Node KR-ACC-72 is powered on and operational.

- Check the QoS policy configuration on the Huawei Access Node using the 'display qos policy' command.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy	High
Inadequate interface queue configuration	Medium
Hardware limitations of the Huawei Access Node	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: QoS policy configuration details, Description: This command displays the QoS policy configuration, including the class maps, policy maps, and service policies. It helps to identify any misconfigured QoS policy that may be causing the congestion event.
```

```
Command: display interface queue, Expected Output: Interface queue configuration details, Description: This command displays the interface queue configuration, including the queue depth, queue size, and other relevant parameters. It helps to identify any inadequate interface queue configuration that may be contributing to the congestion event.

Command: display forwarding table, Expected Output: Forwarding table details, Description: This command displays the forwarding table, including the forwarding entries and their associated QoS policies. It helps to identify any forwarding table inconsistencies that may be causing the congestion event.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: undo qos policy , Description: Use this command to undo the QoS policy configuration and restore the default configuration. Then, reconfigure the QoS policy with the correct settings to prevent future occurrences.

Step 2: Step: Adjust interface queue configuration, Command: queue queue-depth queue-size , Description: Use this command to adjust the interface queue configuration to optimize it for high-speed interfaces. This may involve increasing the queue depth and queue size to prevent packet loss and congestion.

Step 3: Step: Update forwarding table, Command: forwarding-table update, Description: Use this command to update the forwarding table with the corrected QoS policy configuration. This ensures that the forwarding entries are consistent with the QoS policy configuration.

Post-Check Verification

```
Command: display qos policy, Expected Output: Corrected QoS policy configuration details, Description: Use this command to verify that the QoS policy configuration has been corrected and is operating as expected.

Command: display interface queue, Expected Output: Corrected interface queue configuration details, Description: Use this command to verify that the interface queue configuration has been adjusted and is operating as expected.

Command: display forwarding table, Expected Output: Corrected forwarding table details, Description: Use this command to verify that the forwarding table has been updated and is operating as expected.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo qos policy , Description: Use this command to restore the original QoS policy configuration in case the corrective actions fail.

- Step: Restore original interface queue configuration, Command: undo queue queue-depth queue-size , Description: Use this command to restore the original interface queue configuration in case the corrective actions fail.

- Step: Restore original forwarding table, Command: undo forwarding-table update, Description: Use this command to restore the original forwarding table in case the corrective actions fail.